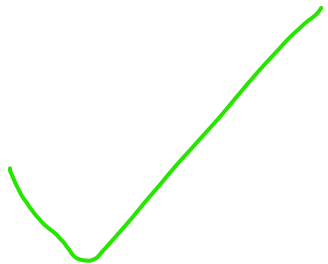


SWI Week 8

⑥

$x_{n+1} =$

$f'''(a)$



✓ If $f(\underline{6}) = 3$, $f'(\underline{6}) = 1/2$ and $f''(\underline{6}) = 2$, what could be the value of $f(7)$?

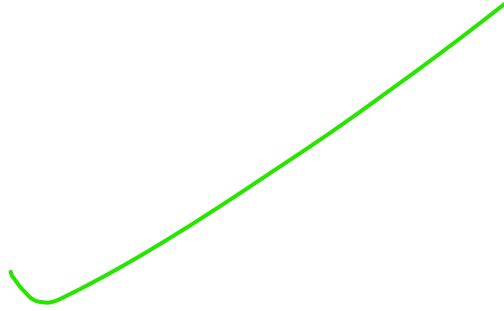
$$f(x+nd) = f(x) + nd f'(x) + \frac{n^2 d^2}{2!} f''(x) + \dots$$

Taylor series approximation.



Students, enter a number!

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$f(x, y) = xy^2$. Take $\eta = 1$ and $(x_0, y_0) = (2, 2)$. What will be the value of the function after two iterations as per Gradient descent?



Students, enter a number!

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4.24, -1.8,

$f(x, y) = 2xy + 2x - x^2 - 2y^2$. Take $\eta = 0.2$ and $(x_0, y_0) = (-1, -1)$. What will be the value of x_2 as per Gradient descent?

$f(x_2, y_2)$



Students, write your response!



At what value of x will $f(x) = (2x - 8)^2$ be minimum?



$$\frac{f''(x) = 8 > 0}{\rightarrow \text{min}}$$

$$f'(x) = 0$$

$$8(x-4) = 0$$

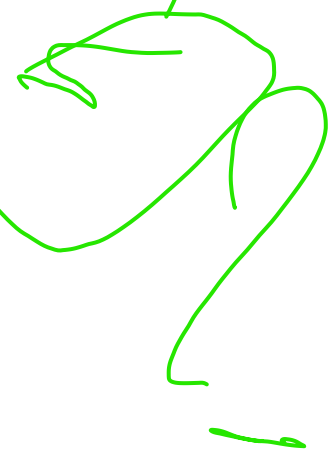
mining

$$x = 4$$



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Range

$$f(x) = (2x - 8)^2$$

↑ ↘

(x=4)

of iterations =

Assuming $x_0 = 2$ and $\eta = 0.05$ What is the number of iterations required to reach (x) in the previous question using the method of Gradient descent? (Consider an error of 0.1)

error = current point - optimal point

$$| \text{error} | \leq 0.1$$

$|x_k - 4| \leq 0.1$



Students, enter a number!

second derivative

Newton $\rightarrow$ Gradient
first derivative

We can also use Newton's method for such optimization. The update rule for Newton's method is given by:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)} \rightarrow \text{Taylor series } \underline{\underline{\text{Approximation}}} \rightarrow \underline{\underline{f''(x)}}$$

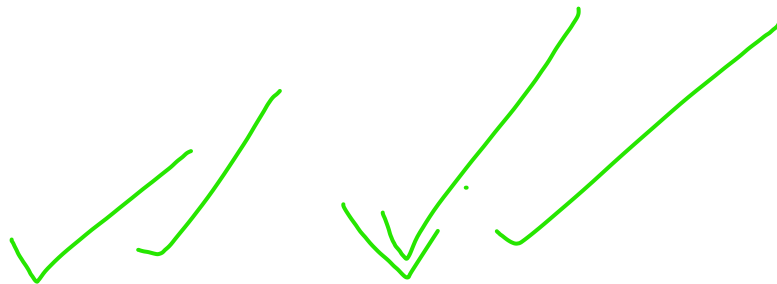
Find the number of iterations required to reach x obtained in Q1 (Consider an error of 0.1).

$$\underline{f(b) = 3}, \quad \underline{f'(b) = 1/2}, \quad \underline{f''(b) = 2}$$

$$f'''(b) + f'''(b) + \dots$$



Students, write your response!



Consider the function $f(x) = x^3 - x^2 - x + 5$,
At which positive value of x is $f(x)$ minimum?



=

$f'(x)$



Students, enter a number!

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Assuming initial guess $x_0 = 0.75$ and Step size, $\eta = 0.25$, What is the number of iterations required to reach x , in the previous question using the method of Gradient descent? (Consider an error of 0.001)



Students, enter a number!

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We can also use Newton's method for such optimization. The update rule for Newton's method is given by:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Find the number of iterations required to reach x obtained in Q1 (Consider an error of 0.001).

$$x_1 = \underline{\underline{5.75}} \quad | \quad 5.75 - 5.75 | = 0 \leq 0.001$$



Students, enter a number!

